KAIZEN MATHS

Lesson starter

SUVAT and Equations of Motion

SUVAT Basics

Name _____

S1

A particle starts with speed 9 m/s and accelerates at 2 m/s² for 6 s. Find its final velocity.

S2

A particle has initial velocity 2 m/s and acceleration 3 m/s². Find the displacement after 4 s.

S3

A particle's velocity increases from 12 m/s to 24 m/s in 6 s. Find its acceleration.

S4

A particle accelerates uniformly from 11 m/s to 39 m/s at 4 m/s². Find the time taken.

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KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

SUVAT Basics

Name _____

E1

A particle starts with speed 13 m/s and accelerates at 5 m/s² for 7 s. Find its final velocity.

E2

A particle has initial velocity 12 m/s and acceleration 4 m/s². Find the displacement after 4 s.

E3

A particle's velocity increases from 8 m/s to 28 m/s in 5 s. Find its acceleration.

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Teacher answers & working

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S1

A particle starts with speed 9 m/s and accelerates at 2 m/s² for 6 s. Find its final velocity.

Answer:

$v = 21\ m/s$

Working:

List the known values.

$u = 9\ m/s,\ a = 2\ m/s^2,\ t = 6\ s,\ v = ?$

Choose the equation containing u, a, t and v.

$v = u + at$

Substitute the values.

$v = 9 + 2 \times 6$

$v = 9 + 12 = 21$

The final velocity is 21 m/s.

S2

A particle has initial velocity 2 m/s and acceleration 3 m/s². Find the displacement after 4 s.

Answer:

$s = 32\ m$

Working:

List the known values.

$u = 2\ m/s,\ a = 3\ m/s^2,\ t = 4\ s,\ s = ?$

Choose the equation containing s, u, a and t.

$s = ut + \frac{1}{2}at^2$

Substitute and calculate each part.

$s = 2 \times 4 + \frac{1}{2} \times 3 \times 4^2$

$s = 8 + 24 = 32$

The displacement is 32 m.

Teacher answers & working

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S3

A particle's velocity increases from 12 m/s to 24 m/s in 6 s. Find its acceleration.

Answer:

$a = 2\ m/s^2$

Working:

List the known values.

$u = 12\ m/s,\ v = 24\ m/s,\ t = 6\ s,\ a = ?$

Start with v = u + at and rearrange.

$v - u = at$

$a = (v - u) \div t$

Substitute the values.

$a = (24 - 12) \div 6$

$a = 12 \div 6 = 2$

The acceleration is 2 m/s².

S4

A particle accelerates uniformly from 11 m/s to 39 m/s at 4 m/s². Find the time taken.

Answer:

$t = 7\ s$

Working:

List the known values.

$u = 11\ m/s,\ v = 39\ m/s,\ a = 4\ m/s^2,\ t = ?$

Start with v = u + at and rearrange.

$v - u = at$

$t = (v - u) \div a$

Substitute the values.

$t = (39 - 11) \div 4$

$t = 28 \div 4 = 7$

The time taken is 7 s.

Teacher answers & working

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E1

A particle starts with speed 13 m/s and accelerates at 5 m/s² for 7 s. Find its final velocity.

Answer:

$v = 48$ m/s

Working:

List the known values.

$u = 13$ m/s, $a = 5$ m/s², $t = 7$ s, $v = ?$

Choose the equation containing u, a, t and v.

$v = u + at$

Substitute the values.

$v = 13 + 5 \times 7$

$v = 13 + 35 = 48$

The final velocity is 48 m/s.

E2

A particle has initial velocity 12 m/s and acceleration 4 m/s². Find the displacement after 4 s.

Answer:

$s = 80$ m

Working:

List the known values.

$u = 12$ m/s, $a = 4$ m/s², $t = 4$ s, $s = ?$

Choose the equation containing s, u, a and t.

$s = ut + \frac{1}{2}at^2$

Substitute and calculate each part.

$s = 12 \times 4 + \frac{1}{2} \times 4 \times 4^2$

$s = 48 + 32 = 80$

The displacement is 80 m.

Teacher answers & working

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E3

A particle's velocity increases from 8 m/s to 28 m/s in 5 s. Find its acceleration.

Answer:

$a = 4\ m/s^2$

Working:

List the known values.

$u = 8\ m/s,\ v = 28\ m/s,\ t = 5\ s,\ a = ?$

Start with v = u + at and rearrange.

$v - u = at$

$a = (v - u) \div t$

Substitute the values.

$a = (28 - 8) \div 5$

$a = 20 \div 5 = 4$

The acceleration is 4 m/s².